

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm. Examination

University Theory Examination January 2018

PH112 HUMAN ANATOMY PHYSIOLOGY I

Date: 02.01.2018, Tuesday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Which of these is **not** a function of the kidneys?
A) The kidneys stimulates the activity of osteoclasts.
B) The kidneys promote RBCs production
C) The kidneys help to regulate blood pressure.
D) The kidneys regulate the volume of body fluids.
 2. Which of the following organ is not the part of urinary system?
A) Ureters
B) Urinary Bladder
C) Kidney
D) Rectum
 3. Which structure is last in the sequence of urine flow?
A) ureter
B) kidney
C) bladder
D) urethra
 4. The outermost structure of a kidney is the _____.
A) cortex
B) medulla
C) capsule
D) pelvis
 5. Which system functions in the excretion of soluble nitrogenous wastes?
A) gastrointestinal
B) pulmonary
C) urinary
D) integumentary
 6. The type of anemia that is fairly common and caused by insufficient dietary iron is _____.
A) aplastic anemia
B) pernicious anemia
C) hemolytic anemia
D) iron deficiency anemia
 7. An acute infection would show up in a blood count as _____.

- A) leukopenia
 - B) leukocytosis
 - C) too few helper T cells
 - D) thrombocytopenia
8. Which blood cell can be described as being a biconcave disc?
- A) platelet
 - B) neutrophil
 - C) eosinophil
 - D) erythrocyte
9. Antibodies are defense substances produced by the _____ cells.
- A) T-lymphocyte
 - B) monocyte
 - C) B-lymphocyte
 - D) leukocyte
10. Cardiac output can be determined by the following formula:
- A) $HR - SV$
 - B) $HR \text{ divided by } SV$
 - C) $HR + SV$
 - D) $HR \times SV$
11. The first in the cardiac conduction sequence is the _____.
- A) SA node
 - B) AV node
 - C) AV bundle
 - D) Purkinje fibers
12. The P wave of an electrocardiogram represents which event?
- A) ventricular depolarization
 - B) atrial repolarization
 - C) ventricular repolarization
 - D) atrial depolarization
13. A heartbeat of less than 60 beats per minute is called _____.
- A) tachycardia
 - B) bradycardia
 - C) arrhythmia
 - D) fibrillation
14. Intrinsic factor is secreted by which cells of the gastric glands?
- A) mucous cells
 - B) chief cells
 - C) peptic cells
 - D) parietal cells
15. The enzyme salivary amylase begins to digest which molecules?
- A) proteins
 - B) carbohydrates
 - C) lipids
 - D) nucleic acids
16. Where would the least amount of bacteria be found?
- A) small intestine
 - B) colon
 - C) stomach
 - D) pharynx
17. The exchange of gases between the blood and the cells of the body is called _____.
- A) external respiration
 - B) internal respiration
 - C) pulmonary ventilation

- D) cellular respiration
18. What is the last structure in this respiratory sequence?
- A) lungs
B) trachea
C) larynx
D) pharynx
19. The only structures that allow gas diffusion across them are the ____.
- A) bronchioles
B) alveoli
C) alveolar ducts
D) bronchi
20. Which of the following is respiratory system disorder?
- A) Tuberculosis
B) Peptic ulcer
C) Hepatitis
D) Leukemia

SECTION – II

- Q 2** Attempt any TWO of the following
- A** Discuss the physiology of urine formation. 05
- B** Write a note on anatomy of lungs. 05
- C** Draw a neat and labeled diagram of nephron. 05
- Q 3** Attempt any TWO of the following
- A** Discuss mechanism of respiration. 05
- B** Write a note on life cycle of RBCs. 05
- C** Classify WBCs. Discuss physiology of WBCs. 05

SECTION – III

- Q 4** Attempt any FOUR of the following
- A** Write a note on Blood Clotting mechanism. 05
- B** Explain Electrocardiogram (ECG) in detail 05
- C** Write down functions of blood in detail. 05
- D** Explain following terms: 05
- 1) Constipation 2) Hypertension 3) Color index
4) Erythropoiesis 5) Angina
- E** Draw labelled diagram of 1) Heart 2) Digestive system 05
- F** Explain briefly about Blood pressure & Write a note on RAS. 05
- Q 5** Attempt any FOUR of the following
- A** Explain physiology of gastric acid secretion. 05
- B** Write a note on Cardiac Cycle. 05
- C** What do you mean by cardiac output? Explain factor affecting cardiac output. 05
- D** Explain functions of liver in detail. 05
- E** What is anaemia? Classify and explain different types of anaemia. 05
- F** Write a note on Pancreas. 05